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7 September 2021

10 September 2022

30 June 2023

12 August 2023

15 March 2024

22 May 2025

Chlor吡格雷 Sulfate Tablets Instructions

Read the instructions carefully and use under the guidance of a physician or pharmacist.

Drug Name

Generic Name: Clopidogrel Hydrogen sulfate Tablets

Product Name: Plavix® PLAVIX®

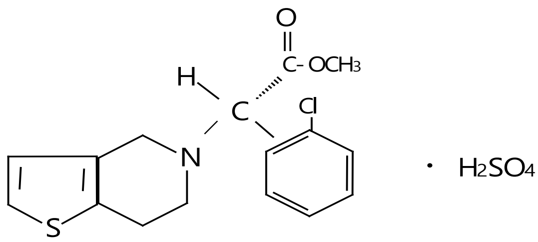
English name: Clopidogrel Bisulfate Tablets

Pinyin: Liusuanqinglubigelei Pian

[Ingredients]

Chemical name: S(+)-2-(2-chlorophenyl)-2-(4,5,6,7-tetrahydrothiophen 并 [3,2-c] pyridin-5-yl)acetate sulfate

chemical structural formula：



Molecular formula: $C_{16}H_{16}ClNO_2S \cdot H_2SO_4$

Molecular weight: 419.90

Excipients: Mannitol, Microcrystalline cellulose, Polyethylene glycol 6000, Low-substituted hydroxypropyl cellulose, Hydrogenated castor oil, Film-coating premix, Brazilian palm wax.

【 Shape and Properties 】

This product is a pink, round film-coated tablet that appears white or off-white after removal of the coating.

【 Indication 】

Clopidogrel is indicated for secondary prevention of atherosclerotic thrombotic events in the following patients:

- Patients with recent myocardial infarction (ranging from several days to less than 35 days), recent ischemic stroke (ranging from 7 days to less than 6 months), or those diagnosed with peripheral arterial disease.
- Patients with acute coronary syndrome

-For non-ST-segment elevation acute coronary syndrome (including unstable angina or non-Q-wave myocardial infarction), particularly in patients who have undergone percutaneous coronary intervention with stent placement, administer aspirin in combination.

For patients with ST-segment elevation acute coronary syndrome, it can be used in combination with aspirin during percutaneous coronary intervention (including patients receiving post-procedural stent placement) or during thrombolytic therapy.

For more information, refer to [Clinical Trial].

【 Specifications 】

75 mg (calculated based on C16H16ClNO2S)

【 Usage and Dosage 】

- Adults and the elderly

The recommended dosage of clopidogrel is 75 mg once daily. Administer orally, with or without food.

For patients with acute coronary syndrome:

For patients with non-ST-segment elevation acute coronary syndrome (unstable angina or non-Q-wave myocardial infarction), treatment should be initiated with a single loading dose of clopidogrel 300 mg or 600 mg. Patients aged <75 years may consider receiving a 600 mg loading dose prior to planned percutaneous coronary intervention (see [Precautions]). Subsequent clopidogrel therapy should be administered continuously at 75 mg once daily (combined with aspirin 75 mg–325 mg/day). Due to the higher bleeding risk associated with higher aspirin doses, the recommended daily maintenance dose of aspirin should not exceed 100 mg. The optimal treatment duration has not been formally established; clinical trial data support a 12-month regimen, with maximal efficacy observed after 3 months of therapy (see [Clinical Trials]).

-ST-segment elevation acute myocardial infarction (STEMI):

For patients receiving thrombolytic therapy, initiation should be with a loading dose of 300 mg clopidogrel, followed by 75 mg once daily in combination with aspirin; thrombolytics may or may not be administered concomitantly. A loading dose of clopidogrel is not recommended for patients aged over 75 years. Combination therapy should be initiated as early as possible after symptom onset and continued for at least 4 weeks. Currently, no studies have confirmed the benefits of combining clopidogrel and aspirin for more than 4 weeks (see [Clinical Trials]).

– Percutaneous coronary intervention (PCI) is planned:

- Patients undergoing direct percutaneous coronary intervention (PCI) or PCI performed more than 24 hours after thrombolytic therapy should receive a loading dose of clopidogrel 600 mg. For patients aged ≥75 years, an appropriate loading dose should be selected after thorough assessment of bleeding risk (see [Precautions]).

- Patients who undergo percutaneous coronary intervention within 24 hours after thrombolytic therapy should receive a loading dose of clopidogrel 300 mg.

Subsequent clopidogrel therapy should be administered at 75 mg once daily in combination with aspirin at 75 mg to 100 mg daily. Combined therapy should be initiated as early as possible after symptom onset and continued for 12 months (see [Clinical Trials]).

For patients with recent myocardial infarction (ranging from several days to less than 35 days), recent ischemic stroke (ranging from 7 days to less than 6 months), or diagnosed peripheral arterial disease: the recommended dose is 75 mg daily.

- In case of missed dose:

  -Missed dose within 12 hours of the scheduled medication time: The patient should immediately take one standard dose as a make-up dose and administer the next dose at the scheduled time.

  -Missed dose more than 12 hours after the scheduled medication time: The patient should take the standard dose at the next scheduled medication time without doubling the dosage.

- Children and minors:

  18 The safety and efficacy in pediatric patients have not been established.

- Renal function impairment:

  There is limited treatment experience for patients with renal injury (see [Precautions]).

- Liver function impairment:

  There is limited treatment experience for patients with moderate liver injury who have a bleeding tendency (see [Precautions]).

【 Untoward Effect 】

The safety of clopidogrel has been evaluated in over 44,000 patients, with 12,000 patients receiving treatment for at least 1 year. Clinically relevant adverse reactions observed in the CAPRIE, CURE, CLARITY, COMMIT, and ACTIVE-A studies will be discussed below. In the CAPRIE study, clopidogrel 75 mg/day demonstrated better tolerability compared to aspirin 325 mg/day. Overall, clopidogrel's tolerability was similar to that of aspirin and was independent of age, gender, or ethnicity. In addition to clinical study data, spontaneous reports of adverse reactions were also available.

In clinical studies and post-marketing reports, hemorrhage is the most common adverse reaction, typically reported within the first month of treatment.

In the CAPRIE study, the overall incidence of bleeding events was 9.3% among patients treated with clopidogrel or aspirin. The incidence of serious events with clopidogrel was similar to that with aspirin.

In the CURE study, patients who discontinued medication for more than 5 days prior to surgery exhibited a lower incidence of major bleeding within 7 days post-coronary artery bypass grafting (CABG). Among patients who continued treatment within 5 days after the bypass procedure, the event rates were 9.6% with clopidogrel + aspirin and 6.3% with placebo + aspirin.

In the CLARITY study, the clopidogrel + aspirin group exhibited increased bleeding compared to the placebo + aspirin group. The incidence of major bleeding was similar between the two groups. This finding remained consistent across subgroups stratified by baseline characteristics, type of fibrinolytic agent, or presence/absence of heparin therapy.

In the COMMIT study, the overall incidence rates of intracranial hemorrhage and non-intracranial major hemorrhage were relatively low and similar between the two groups.

In the ACTIVE-A trial, the incidence of major bleeding was higher in the clopidogrel + ASA group compared to the placebo + ASA group (6.7% vs. 4.3%). Major bleeding occurred predominantly in extracranial tissues in both groups (5.3% in the clopidogrel + ASA group vs. 3.5% in the placebo + ASA group), with gastrointestinal bleeding being the most common manifestation (3.5% vs. 1.8%). Intracranial hemorrhage was more frequent in the clopidogrel + ASA group (1.4% vs. 0.8%). No statistically significant differences were observed between the two groups regarding the incidence of fatal bleeding (1.1% vs. 0.7% in the clopidogrel + ASA group versus 0.8% vs. 0.6% in the placebo + ASA group) or the incidence of hemorrhagic stroke.

The adverse reactions reported in clinical studies and spontaneous reports are listed in the table below. The frequency of adverse reactions is defined as follows: Common (≥1/100 to <1/10); Uncommon (≥1/1,000 to <1/100); Rare (≥1/10,000 to <1/1,000); Very rare (<1/10,000); Unknown (无法根据 available data determine). In each system organ classification, adverse reactions are ranked by decreasing severity.

Classification of Organ Syst-ems	Common	Uncommon	Seldom seen	Very rare, unknown*
Abnormalities of the blood and lymphatic systems		Thrombocyto-penia, leukope-nia, eosinophilia	Neutropenia, including sev-ere neutropenia	Thrombotic thro-mbocytopenic pu-rpura (TTP), aplas-tic anemia, pancy-topenia, agranulo-cytosis, severe th-rombocytopenia, acquired hemophi-lia A, granulocyt-openia, anemia
Cardiac anomaly				Allergic Coronary Spasm Syndrome (Vascular Sp-asm)

Classification of Organ Systems	Common	Uncommon	Seldom seen	Very rare, unknown*
				Tonic allergic angina/ painful myocardial infarction): Specifically refers to the allergic reaction induced by clopidogrel.
Immunological dysfunction				Serum sickness, allergic reactions, and thienopyridines exhibit cross-allergic reactions (e.g., ticlopidine and prasugrel) (see [Precautions])*. Insulin autoimmune syndrome can lead to severe hypoglycemia, particularly in patients with the HLADRA4 subtype (which is more common in the Japanese population).
Insanity				Hallucinations, confusion of consciousness
Neurological abnormalities		Intracranial hemorrhage (some reports indicate fatal outcomes), headache, paresthesia, dizziness		Taste disorder, loss of taste
Ocular abnormalities		Eye bleeding (conjunctiva, eye, retina)		
Abnormalities of the ear and labyrinth			Vertigo	
Vascular abnormalities	Hematoncus			Severe hemorrhage, surgical wound bleeding, vasculitis, hypotension
Abnormalities of the respiratory system, chest, and mediastinum	Hemorrhinia			Respiratory hemorrhage (hemoptysis, pulmonary hemorrhage), bronchospasm, interstitial pneumonia, eosinophilic pneumonia
Gastrointestinal system abnormalities	Gastrointestinal bleeding, diarrhea, abdominal pain, dyspepsia	Gastric ulcer and duodenal ulcer, gastritis, vomiting, nausea, constipation, gastrointestinal bloating	Retroperitoneal hamorrhage	Fatal gastrointestinal and retroperitoneal hemorrhage, pancreatitis, colitis (including ulcerative or lymphocytic colitis), stomatitis
Abnormalities of the hepatobiliary system				Acute liver failure, hepatitis, abnormal liver function tests
Abnormalities of the skin and subcutaneous tissue	Bruising	Rash, pruritus, cutaneous hemorrhage (purpura)		Bullous dermatitis (toxic epidermal necrolysis, Stevens-Johnson syndrome, erythema multiforme, acute generalized)

Classification of Organ Systems	Common	Uncommon	Seldom seen	Very rare, unknown*
				Erythematous gestational papulopustular eruption (AGEP), vascular hematoma, drug-induced hypersensitivity reaction syndrome, drug eruption with eosinophilia and systemic symptoms syndrome (DRESS), urticaria, eczema, erythema or exfoliative rash, lichen planus
Abnormalities of the musculoskeletal and connective tissues				Skeletal muscle hemorrhage (arthralgia hematoma), arthritis, arthralgia, myalgia
Abnormalities of the kidneys and urinary system		Hematuria		Glomerulonephritis with elevated serum creatinine
Abnormalities of the reproductive system and mammary glands			Gynaecomastia	
Systemic diseases and various reactions at the site of administration	Injection site hemorrhage			Fever
Check up		Prolonged bleeding time, neutropenia, thrombocytopenia		

*The information regarding clopidogrel administration is "unknown."

【 Taboo 】

1. Allergy to the active substance or any component of this product.
2. Severe liver damage.
3. Active pathological bleeding, such as peptic ulcer or intracranial hemorrhage.

【 Matters Need Attention 】

Hemorrhage and hematological abnormalities

Due to the risks of bleeding and hematologic adverse reactions, immediate consideration should be given for complete blood count (CBC) and/or other appropriate tests upon the appearance of clinical signs of bleeding during treatment. Like other antiplatelet agents, clopidogrel should be used with caution in patients whose bleeding risk is increased by trauma, surgery, or other pathological conditions, as well as those receiving aspirin, nonsteroidal anti-inflammatory drugs (NSAIDs)—including Cox-2 inhibitors, heparins, platelet glycoprotein IIb/IIIa (GP IIb/IIIa) antagonists, selective serotonin reuptake inhibitors (SSRIs), serotonin norepinephrine reuptake inhibitors (SNRIs), or thrombolytic agents. These patients require close monitoring for any signs of bleeding, including occult bleeding, particularly during the initial weeks of treatment and/or following cardiac intervention or surgery. Concomitant use of clopidogrel with warfarin is not recommended due to the potential exacerbation of bleeding.

In patients requiring elective surgery, if antiplatelet therapy is not mandatory, clopidogrel should be discontinued 7 days prior to the procedure. Patients must inform their physician of any clopidogrel use before scheduling any surgical intervention or taking any new medications. Clopidogrel prolongs bleeding time and should be used with caution in patients with bleeding disorders (particularly gastrointestinal or ocular conditions).

Patients should be informed that the hemostasis time may be prolonged when taking clopidogrel (either alone or in combination with aspirin), and they should report any abnormal bleeding events (including location and timing) to their physician.

In patients with non-ST-segment elevation acute coronary syndrome (NSTEMCS) aged ≥75 years, the use of a 600 mg loading dose of clopidogrel is not recommended due to limited data and the increased bleeding risk in this population.

Due to limited clinical data available for STEMI PCI patients aged ≥75 years and the increased bleeding risk, the use of a 600 mg loading dose of clopidogrel should only be considered after individualized assessment of the patient's bleeding risk by the physician.

drug withdrawal

Treatment interruption should be avoided. If clopidogrel discontinuation is necessary, therapy should be resumed as early as possible. Premature discontinuation of clopidogrel may increase the risk of cardiovascular events.

Thrombotic thrombocytopenic purpura (TTP)

Thrombotic thrombocytopenic purpura (TTP) rarely occurs after clopidogrel administration, sometimes manifesting shortly (<2 weeks) post-treatment. TTP can be life-threatening and is characterized by thrombocytopenia, microangiopathic hemolytic anemia, accompanied by neurological abnormalities, renal impairment, or fever. TTP constitutes a medical emergency requiring urgent intervention, including plasma exchange therapy.

Recent ischemic stroke

Due to insufficient data, clopidogrel is not recommended within 7 days after an acute ischemic stroke. Acquired hemophilia

After administration of clopidogrel, cases of acquired hemophilia have been reported. Acquired hemophilia should be considered when there is a definite prolongation of the activated partial thromboplastin time (aPTT), with or without bleeding. Patients diagnosed with acquired hemophilia should be managed and treated by a specialist, and clopidogrel should be discontinued.

Cytochrome P450 2C19 (CYP2C19)

Genetic Pharmacology: In individuals with slow CYP2C19 metabolism, the plasma concentrations of active metabolites of clopidogrel at the recommended dosage are reduced, leading to diminished antiplatelet efficacy. Current methods are available for detecting the patient's CYP2C19 genotype.

Since clopidogrel is partially metabolized by CYP2C19 into its active metabolite, administration of drugs that inhibit this enzyme activity may reduce the conversion of clopidogrel to its active metabolite. The clinical relevance of drug interactions remains uncertain. The combination use of potent or moderate CYP2C19 inhibitors is not recommended.

The use of drugs that induce CYP2C19 activity is expected to increase plasma concentrations of active clopidogrel metabolites and may elevate the risk of bleeding. As a precautionary measure, concomitant use with potent CYP2C19 inducers is not recommended (see [Drug Interactions]).

In patients with recent transient ischemic events or ischemic stroke who have a higher risk of recurrent ischemic events, the combination of aspirin and clopidogrel did not demonstrate greater efficacy than clopidogrel alone, although it increased the risk of bleeding.

Cross-allergic reaction with thiophenopyridine

Thiophenopyridines may cause mild to severe allergic reactions, such as rash, angioedema, or hematologic adverse reactions, including thrombocytopenia and neutropenia. Given existing reports of cross-allergic reactions among thiophenopyridines, patients' allergy histories to other thiophenopyridines (e.g., ticlopidine and prasugrel) should be evaluated (see [Adverse Reactions]). Patients with a history of allergic reactions and/or hematologic adverse reactions to one thiophenopyridine may have an increased risk of experiencing the same or other adverse reactions to another thiophenopyridine. Cross-reactivity monitoring is recommended.

Renal impairment

The experience with clopidogrel use in patients with renal impairment is limited; therefore, clopidogrel should be administered with caution in these patients. Hepatic impairment

For patients with moderate liver disease who may have a bleeding tendency, clopidogrel should be used with caution in such cases due to limited experience with its administration in this population.

Excipient

Plavix contains lactose; it should not be used by patients with rare genetic disorders such as galactose intolerance, Lapp lactase deficiency, or glucose-galactose malabsorption.

This medication contains hydrogenated castor oil, which may cause gastric discomfort and diarrhea.

No effects on driving or machinery operation were observed after administration of clopidogrel.

[Use in Pregnant and Lactating Women]

- period of pregnancy

Since there is currently no clinically available data on the use of clopidogrel during pregnancy, clopidogrel should be avoided in pregnant women as a precautionary measure. Animal studies have provided no direct or indirect evidence indicating that clopidogrel has adverse effects on pregnancy, embryonic/fetal development, delivery, or postnatal growth (see [Pharmacology and Toxicology]).

- suckling period

Animal experimental results indicate that clopidogrel and/or its metabolites can be excreted in breast milk, but it is unclear whether the drug is secreted in human breast milk. As a precaution, breastfeeding should be discontinued during clopidogrel treatment.

- Fertility

No alterations in reproductive function were observed with clopidogrel in animal experiments.

[Pediatric Medication]

There is no experience with its use in children.

[Geriatric Medication]

Refer to [Dosage and Administration].

【 Drug Interaction 】

Drugs associated with bleeding risk: The risk of bleeding increases due to potential cumulative effects. Caution should be exercised when combining clopidogrel with these medications.

Oral anticoagulants: Due to the potential increase in bleeding risk, concomitant use of clopidogrel with oral anticoagulants is not recommended (see [Precautions]). Although daily administration of 75 mg clopidogrel does not alter the outcomes in patients receiving long-term warfarin therapy.

The pharmacokinetics of S-warfarin or its International Normalized Ratio (INR) in patients indicate that the combined use of warfarin and clopidogrel increases the risk of bleeding due to their independent inhibition of the hemostatic process.

Glycoprotein IIb/IIIa antagonists: Caution should be exercised when combining clopidogrel with glycoprotein IIb/IIIa antagonists.

Acetylsalicylic acid (aspirin): Aspirin does not alter the inhibitory effect of clopidogrel on ADP-induced platelet aggregation, but clopidogrel enhances the inhibitory effect of aspirin on collagen-induced platelet aggregation. However, concomitant administration of aspirin 500 mg twice daily for one day did not significantly increase the bleeding time prolongation induced by clopidogrel. A pharmacodynamic interaction between clopidogrel and aspirin may occur, potentially increasing the bleeding risk; therefore, close monitoring is required when the two drugs are used together (see [Precautions]). Nevertheless, cases of concomitant use of clopidogrel and aspirin for more than one year have been reported (see [Clinical Trials]).

Heparin: Studies conducted in healthy volunteers have demonstrated that clopidogrel does not alter the anticoagulant effect of heparin, and no adjustment of heparin dosage is required. Concomitant use of heparin does not affect the platelet aggregation-inhibiting effect of clopidogrel. A pharmacodynamic interaction between clopidogrel and heparin may occur, potentially increasing the risk of bleeding; therefore, close monitoring is advised when these drugs are used together (see [Precautions]).

Thrombolytic agents: In patients with acute myocardial infarction, the safety of combining clopidogrel with fibrin-specific or non-specific thrombolytics and heparin has been evaluated. The incidence of clinical bleeding was similar among those receiving combinations of thrombolytics, heparin, and aspirin (see [Adverse Reactions]).

Nonsteroidal anti-inflammatory drugs (NSAIDs): In a clinical trial conducted among healthy volunteers, concomitant use of clopidogrel and naproxen increased the incidence of occult gastrointestinal bleeding. Due to the lack of research on interactions between clopidogrel and other NSAIDs, it remains unclear whether concomitant use with all NSAIDs elevates the risk of gastrointestinal bleeding events. Therefore, caution is advised when combining NSAIDs—including Cox-2 inhibitors—with clopidogrel (see [Precautions]).

Selective serotonin reuptake inhibitors (SSRIs) and selective norepinephrine reuptake inhibitors (SNRIs): Due to their effect on platelet activation, concomitant use of SSRIs/SNRIs with clopidogrel may increase the risk of bleeding.

Other combination therapies

CYP2C19 inducer

Since clopidogrel is partially metabolized by CYP2C19 into its active metabolite, the use of drugs that induce this enzyme activity is expected to increase the plasma concentrations of clopidogrel's active metabolite.

Rifampin strongly induces CYP2C19, leading to increased levels of active metabolites of clopidogrel and platelet inhibition, particularly potentially elevating the risk of bleeding. As a precautionary measure, concomitant use with potent CYP2C19 inducers is not recommended (see [Precautions]).

CYP2C19 inhibitor

Since clopidogrel is partially metabolized by CYP2C19 into its active metabolite, the use of drugs that inhibit this enzyme activity will lead to reduced levels of the active clopidogrel metabolite. The clinical relevance of drug interactions remains uncertain. Concomitant use of potent or moderate CYP2C19 inhibitors (e.g., omeprazole) is not recommended (see [Precautions] and [Clinical Pharmacology]).

Examples of potent or moderate CYP2C19 inhibitors include: omeprazole and esomeprazole, fluvoxamine.

Fluoxetine, moclobemide, voriconazole, fluconazole, ticlopidine, carbamazepine, and efavirenz. Proton pump inhibitors (PPIs).

Omeprazole 80 mg administered once daily, either concomitantly with clopidogrel or at an interval of 12 hours, reduced the plasma concentrations of the active metabolite of clopidogrel by 45% (load dose) and 40% (maintenance dose), respectively. This reduction in plasma concentration led to a corresponding decrease in platelet aggregation inhibition rates by 39% (load dose) and 21% (maintenance dose), respectively. Esomeprazole and clopidogrel may exhibit similar interactions.

Regarding the impact of pharmacokinetic (PK)/pharmacodynamic (PD) interactions on clinical outcomes such as major cardiovascular events, observational and clinical study results are inconsistent. The combination of clopidogrel with omeprazole or esomeprazole is not recommended (see [Precautions]).

No significant reduction in plasma concentrations of clopidogrel metabolites was observed after the combination of pantoprazole, lansoprazole, and clopidogrel.

When administered in combination with pantoprazole 80 mg once daily, the plasma concentrations of the active metabolites of clopidogrel decreased by 20% (load dose) and 14% (maintenance dose), respectively, accompanied by reductions in the average platelet aggregation inhibition rate by 15% and 11%, respectively. These results suggest that clopidogrel can be co-administered with pantoprazole.

No evidence suggests that other gastric acid secretion inhibitors, such as H2-blockers (excluding the CYP2C19 inhibitor cimetidine) or antacids, interfere with the antiplatelet activity of clopidogrel.

Enhanced Antiretroviral Therapy (ART)

HIV patients receiving intensive antiretroviral therapy (ART) face a high risk of vascular events.

In HIV patients receiving ritonavir or cobimetastat-enhanced ART, a significant reduction in platelet inhibition has been observed. Although the clinical relevance of these findings remains uncertain, spontaneous reports have indicated that HIV-infected patients on ritonavir-enhanced antiretroviral therapy experienced recurrent occlusion events after clearance of occlusion or thrombotic events following clopidogrel loading therapy. Concurrent use of clopidogrel with ritonavir reduces the average degree of platelet inhibition. Therefore, concomitant use of clopidogrel with enhanced antiretroviral therapy is not recommended.

Other medications

Through extensive clinical studies, the pharmacodynamic and pharmacokinetic interactions of clopidogrel with other concomitant medications have been investigated. No clinically significant pharmacodynamic interactions were observed when clopidogrel was administered alone or in combination with atenolol or nifedipine. Additionally, concomitant use of clopidogrel with phenobarbital or estradiol did not significantly affect its pharmacodynamic activity.

Clopidogrel does not alter the pharmacokinetics of digoxin or theophylline. Antacids do not affect the absorption of clopidogrel.

The CAPRIE study data indicate that phenytoin and tolbutamide can be safely co-administered with clopidogrel.

CYP2C8 substrate drugs: Clopidogrel increases the exposure of repaglinide in healthy volunteers. In vitro studies have demonstrated that the increased repaglinide exposure is attributed to the inhibition of CYP2C8 by clopidogrel glucuronide metabolites. Due to the risk of elevated plasma drug concentrations, caution should be exercised when combining clopidogrel with drugs primarily metabolized and cleared via CYP2C8 (e.g., repaglinide, paclitaxel).

In addition to the aforementioned clearly documented drug interaction information, no studies have investigated the interactions between commonly used medications in patients with atherosclerotic thrombotic diseases and clopidogrel. However, during clinical trials, patients receiving clopidogrel concurrently were administered multiple concomitant drugs, including diuretics, beta-blockers, ACE inhibitors, calcium antagonists, lipid-lowering agents, coronary dilators, antidiabetic agents (including insulin), antiepileptic drugs, and GPⅡb/Ⅲa receptor antagonists, with no clinically significant adverse interactions observed.

Like other oral P2Y12 inhibitors, concomitant use of opioid agonists may delay and reduce the absorption of clopidogrel, likely due to slowed gastric emptying. The clinical relevance remains unclear. In patients with acute coronary syndrome requiring concurrent administration of morphine or other opioid agonists, parenteral antiplatelet agents may be considered.

Rosuvastatin: After administration of 300 mg clopidogrel, clopidogrel increased the rosuvastatin exposure AUC by 2-fold and Cmax by 1.3-fold; after 75 mg clopidogrel, it increased the rosuvastatin exposure AUC by 1.4-fold but had no effect on Cmax.

【 Overdose 】

Overdose of clopidogrel may lead to prolonged bleeding time and bleeding complications. Appropriate management should be implemented if bleeding is observed.

No antidote has been identified for the pharmacological activity of clopidogrel. If prolonged bleeding time requires prompt correction, platelet infusion can reverse the effects of clopidogrel.

[Clinical Pharmacology]

mechanism of action

Clopidogrel is a prodrug, and one of its metabolites can inhibit platelet aggregation. Clopidogrel must be metabolized by the CYP450 enzyme to generate active metabolites that inhibit platelet aggregation. The active metabolites of clopidogrel selectively inhibit the binding of adenosine diphosphate (ADP) to platelet P2Y12 receptors and the subsequent ADP-mediated activation of the glycoprotein GPⅡb/Ⅲa complex, thereby suppressing platelet aggregation. Due to the irreversible nature of this binding, the residual lifespan of platelets exposed to clopidogrel (approximately 7–10 days) is affected, while the recovery rate of normal platelet function aligns with platelet renewal. By blocking the ADP-induced pathway of platelet activation and aggregation, other agonists besides ADP can also be inhibited in their platelet-suppressing effects.

Since active metabolites are formed via CYP450 enzymes, and some CYP450 enzymes are polymorphic or inhibited by other drugs, not all patients will achieve adequate platelet inhibition.

pharmacodynamics

Clopidogrel 75 mg, administered once daily, significantly inhibited ADP-induced platelet aggregation from day one; the inhibitory effect gradually intensified and reached steady state within 3 to 7 days. At steady state, the average inhibition level with daily administration of clopidogrel 75 mg ranged from 40% to 60%. Platelet aggregation and bleeding time typically returned to baseline levels within 5 days after discontinuation of treatment.

pharmacokinetics

absorb

A single or multiple oral dose of 75 mg clopidogrel is rapidly absorbed. The average plasma concentration of the parent compound clopidogrel peaks approximately 45 minutes after administration (approximately 2.2–2.5 ng/mL following a single oral dose of 75 mg). Based on the excretion of clopidogrel metabolites in urine, at least 50% of the drug is absorbed.

distribution

In vitro studies demonstrate that clopidogrel and its primary circulating metabolite (inactive) exhibit reversible binding to human plasma proteins (98% and 94%, respectively), remaining in a non-saturated state across a wide concentration range.

supersession

Clopidogrel is primarily metabolized in the liver. Its in vivo and in vitro metabolism occurs through two main metabolic pathways: one pathway mediated by esterase, where hydrolysis converts it into inactive acid derivatives (85%).

One metabolic pathway involves the conversion of the drug into intermediate metabolites, while the other pathway is mediated by multiple cytochrome P450 enzymes. Clopidogrel is first metabolized into the intermediate metabolite 2-oxyclopidogrel. This intermediate is subsequently converted into the active metabolite, a clopidogrel thiol derivative. The active metabolite is primarily produced through the action of CYP2C19 and several other CYP enzymes, including CYP3A4, CYP1A2, and CYP2B6. The isolated active thiol derivative binds rapidly and irreversibly to platelet receptors in vitro, thereby inhibiting platelet aggregation.

The Cmax of the active metabolite following a single 300 mg clopidogrel loading dose was twice that observed after four days of maintenance dosing. The peak plasma concentration (Cmax) was achieved approximately between 30 to 60 minutes post-administration.

eliminate

After oral administration of ^{14}C-labeled clopidogrel, approximately 50% is excreted in urine and about 46% in feces within 120 hours. Following a single oral dose of 75 mg clopidogrel, the half-life of clopidogrel is 6 hours, while the half-life of its active metabolite is approximately 30 minutes. Following both single and repeated administration, the elimination half-life of the inactive metabolites (without biological activity) in circulation is 8 hours.

pharmacogenetics

CYP2C19 is involved in the formation of active metabolites and 2-oxo-clopidogrel intermediate metabolites. The pharmacokinetics and antiplatelet effects (the latter measured by in vitro platelet aggregation assays) of clopidogrel active metabolites vary depending on the CYP2C 19 genotype.

The CYP2C19*1 allele corresponds to the fully functional metabolic phenotype, whereas the CYP2C19*2 and CYP2C19*3 alleles are functionally deficient. The CYP2C19*2 and CYP2C 19*3 alleles account for 85% of the slow metabolism alleles in Caucasians and 99% in Asians. Other alleles associated with the slow metabolism phenotype include CYP2C19*4, *5, *6, *7, and *8, though these are less common. Individuals with the slow metabolism phenotype carry two of the aforementioned functionally deficient alleles. The reported distribution frequencies of CYP 2C19 slow metabolism genotypes are approximately 2% in Caucasians, 4% in Africans, and 14% in China. Current methods exist for detecting patient CYP2C19 genotypes.

In a crossover trial conducted involving 40 healthy subjects, four CYP2C19 metabolic phenotype groups (ultrafast metabolizer, fast metabolizer, intermediate metabolizer, and slow metabolizer) were established, with 10 subjects enrolled in each group. The pharmacokinetic characteristics and antiplatelet function of subjects in each group were evaluated. The dosing regimens were as follows: initial dose of 300 mg followed by 75 mg/day; initial dose of 600 mg followed by 150 mg/day; both regimens lasted a total of 5 days (steady state). No significant differences were observed in plasma concentrations of active metabolites or mean platelet aggregation inhibition rate (IPA) among subjects with ultrafast, fast, and intermediate metabolism. Plasma concentrations of active metabolites in slow metabolizers were 63%–71% lower than those in fast metabolizers. After administration of the 300 mg/75 mg regimen, antiplatelet activity was reduced in slow metabolizers, with mean IPA (at 5 μM ADP) of 24% (at 24 hours) and 37% (on day 5), compared to 39% (at 24 hours) and 58% (on day 5) in fast metabolizers, and 37% (at 24 hours) and 60% (on day 5) in intermediate metabolizers. Slow metabolizers receiving the 600 mg/150 mg regimen exhibited higher plasma concentrations of active metabolites than those receiving the 300 mg/75 mg regimen. Furthermore, subjects receiving the 600 mg/150 mg dosing regimen exhibited an IPA of 32% (at 24 hours) and 61% (on day 5), which were higher than those observed in slow代谢 subjects receiving the 300 mg/75 mg regimen. In slow代谢 subjects receiving 600 mg/150 mg, the measured plasma concentrations of active metabolites and IPA values reached levels comparable to those observed in other metabolic subgroups receiving 300 mg/75 mg. Currently, there is a lack of clinical endpoint studies in slow metabolism patients to help determine appropriate dosages and dosing regimens for this population.

A meta-analysis of 335 patients treated with clopidogrel across six studies under steady-state conditions yielded similar results: intermediate metabolizer patients exhibited 28% lower active metabolite exposure compared to fast metabolizers.

The metabolic rate was reduced by 72% in individuals with slow metabolism; simultaneously, platelet aggregation inhibition (induced by 5 μM ADP) was also decreased. The differences in IPA between these individuals and those with fast metabolism were 5.9% and 21.4%, respectively.

There is a lack of prospective, randomized, controlled trial results to evaluate the impact of CYP2C19 genotypes on clinical outcomes in patients receiving clopidogrel therapy. However, several retrospective analyses have assessed changes in clinical outcomes among patients with different genotypes after clopidogrel treatment: CURE (n = 2,721), CHARISMA (n = 2,428), CLARITY-TIMI 28 (n = 227), TRITON-TIMI 38 (n = 1,477), and ACTIVEA (n = 601); in addition, several published cohort studies are available.

In the TRITON-TIMI 38 and three cohort studies (Collet, Sibbing, Giusti), analyses combining patients with intermediate and slow metabolism showed higher incidences of cardiovascular events (death, myocardial infarction, and stroke) or stent thrombosis compared to patients with fast metabolism.

In both CHARISMA and a cohort study (Simon), an increased incidence of events was observed only in patients with slow metabolism compared to those with fast metabolism.

In the CURE, CLARITY, ACTIVE A, and a cohort study (Trenk), no increased incidence of cardiovascular events was observed among patients with different CYP2C19 metabolicotyp-es.

The sample sizes in these analyses may be insufficient to detect differences in clinical endpoints among patients with a slow metabolic type.

Special populations

The pharmacokinetics of clopidogrel active metabolites in these specific populations remain unclear.

Renal impairment: After repeated administration of clopidogrel 75 mg once daily, patients with severe renal impairment (creatinine clearance 5–15 mL/min) exhibited a lower inhibition of ADP-induced platelet aggregation (25%) compared to healthy subjects; however, the prolongation of bleeding time was comparable to that observed in healthy volunteers receiving clopidogrel 75 mg daily. Additionally, all patients demonstrated good clinical tolerance.

Liver function impairment: In patients with severe liver dysfunction, the inhibitory effect of daily oral clopidogrel 75 mg administered for 10 days on ADP-induced platelet aggregation was similar to that observed in healthy subjects. The degree of average bleeding time prolongation was also comparable in both groups.

Race: The genotypes associated with intermediate and slow CYP2C19 metabolizers vary across different racial/ethnic groups (see Genetic Pharmacology). According to existing literature, limited reference data are available for assessing the clinical significance of CYP2C 19 genotype in predicting clinical endpoint events among Asian populations.

【 Clinical Test 】

The efficacy and safety of clopidogrel have been evaluated in a total of five double-blind clinical trials involving over 88,000 patients: the CAPRIE study, which compared clopidogrel with aspirin; and the CURE, CLARITY, COMMIT, and ACTIVE-A studies, which compared the efficacy and safety of clopidogrel with placebo against aspirin and other standard therapies.

Recent myocardial infarction (MI), recent stroke, or confirmed peripheral arterial disease

The CAPRIE study enrolled a total of 19,185 patients presenting with recent myocardial infarction (<35 days), recent ischemic stroke (7 days to 6 months), or confirmed peripheral arterial disease (PAD) complicated by atherosclerotic thrombosis. Patients were randomly assigned to receive either clopidogrel 75 mg/day or aspirin 325 mg/day and were followed for 1 to 3 years. In the myocardial infarction subgroup, the majority of patients received aspirin therapy within the first few days of acute myocardial infarction onset.

Compared with aspirin, clopidogrel significantly reduced the incidence of new ischemic events (the combined endpoint included myocardial infarction, ischemic stroke, and vascular death). The intention-to-treat analysis revealed 939 events in the clopidogrel group and 1,020 events in the aspirin group (relative risk reduction [RRR] of 8.7%, [95% CI: 0.2–16.4]; P = 0.045), indicating that for every 1,000 patients treated with clopidogrel for 2 years, 10 additional patients were prevented from developing new ischemic events compared to aspirin [CI: 0–20]. In the analysis using total mortality as the secondary endpoint, no significant difference was observed between the clopidogrel group (5.8%) and the aspirin group (6.0%).

Subgroup analyses of eligible patients (those with myocardial infarction, ischemic stroke, or peripheral arterial disease) revealed that patients enrolled due to peripheral arterial disease (particularly those with a history of myocardial infarction) appeared to benefit most (RRR = 23.7%; CI: 8.9–36.2; P = 0.003), whereas stroke patients showed weaker benefits (RRR = 7.3%; CI: -5.7–18.7; P = 0.258), with no statistically significant difference compared to the aspirin group. Among patients enrolled with only recent myocardial infarction, the clopidogrel group demonstrated slightly inferior numerical results compared to the aspirin group, though the difference was not statistically significant (RRR = -4.0%; CI: -22.5–11.7; P = 0.639). Additionally, age-specific subgroup analyses indicated that the benefit of clopidogrel was lower in patients aged>75 years compared to those aged ≤75 years.

Since the efficacy evaluation in individual subgroups of the CAPRIE study lacks sufficient confidence, it remains unclear whether the observed differences in relative risk reduction among these subgroups are genuine or merely due to chance.

acute coronary syndrome

The CURE study enrolled a total of 12,562 patients with non-ST-segment elevation acute coronary syndrome (unstable angina or non-Q-wave myocardial infarction), presenting with chest pain occurring within 24 hours or symptoms consistent with ischemic disease. Patients were required to exhibit electrocardiographic changes indicative of new ischemic changes or elevated myocardial enzymes, troponin I, or troponin T levels at least twice the upper limit of normal. Patients were randomly assigned to receive clopidogrel (load dose of 300 mg followed by 75 mg/day, N=6,259) or placebo (N=6,303), both combined with aspirin (75–325 mg once daily) and other standard therapies. Treatment was administered for one year. In the CURE study, 823 (6.6%) patients received combined therapy with GPⅡb/Ⅲa receptor antagonists. Heparin was used in over 90% of patients, and the relative bleeding incidence did not differ significantly between the clopidogrel and placebo groups despite combined heparin therapy.

The number of patients who experienced the primary endpoint events [cardiovascular death, myocardial infarction, or stroke] was 582 (9.3%) in the clopidogrel group and 719 (11.4%) in the placebo treatment group. The relative risk reduction in the clopidogrel treatment group was 20% (95% CI: 10%–28%; P = 0.00009): a 17% reduction in patients receiving conservative therapy, a 29% reduction in those undergoing PTCA with or without stent implantation, and a 10% reduction in patients undergoing CABG. New-onset cardiovascular events (the primary endpoint) were prevented, with relative risk reductions of 22% (CI: 8.6, 33.4),32% (CI: 12.8, 46.4),4% (CI: -26.9, 26.7),6% (CI: -33.5, 34.3), and 14% (CI: -31.6, 44.2) during the study periods of 0–1, 1–3, 3–6, 6–9, and 9–12 months, respectively. Therefore, after more than three months of treatment, the observed benefit in the clopidogrel plus aspirin regimen did not increase further, while the bleeding risk remained present (see [Precautions]).

In the CURE study, the use of clopidogrel reduced the need for thrombolytic therapy (RRR=43.3%; CI: 24.3%,57.5%) and GPⅡb/Ⅲa inhibitors (RRR=18.2%; CI: 6.5%,28.3%).

In the clopidogrel treatment group and the placebo treatment group, the number of patients who experienced the combined primary endpoint events (cardiovascular death, myocardial infarction, stroke, or refractory ischemia) was 1,035 (16.5%) and 1,187 (18.8%), respectively. The relative risk reduction in the clopidogrel treatment group was 14%. This benefit was primarily attributed to the reduction in myocardial infarction incidence.

A significant reduction in the incidence rate [287 cases (4.6%) in the clopidogrel group and 363 cases (5.8%) in the placebo group]. No effect on the rehospitalization rate due to unstable angina was observed.

The analysis results from populations with diverse characteristics (e.g., unstable angina or non-Q-wave myocardial infarction, low-to high-risk populations, diabetes, those requiring vascular reconstruction, age, gender, etc.) were consistent with the primary analysis results. Notably, a causal analysis conducted on 2,172 patients who underwent stent placement in the CURE trial (representing 17% of all patients enrolled in the CURE clinical trial) demonstrated a 26.2% reduction in the relative risk of major endpoint events (cardiovascular death, myocardial infarction, stroke) and a 23.9% reduction in the relative risk of secondary endpoint events (cardiovascular death, myocardial infarction, stroke, or refractory ischemia) with clopidogrel compared to placebo. Furthermore, the stent placement subgroup in the CURE study did not indicate any safety concerns with clopidogrel. These findings are consistent with the overall results of the CURE study.

The observed benefits of clopidogrel are independent of other acute and long-term cardiovascular therapies (e.g., heparin/low-molecular-weight heparin, GPⅡb/Ⅲa receptor antagonists, lipid-lowering agents, beta-blockers, and ACE inhibitors). The efficacy of clopidogrel is also independent of the aspirin dosage (75–325 mg/day).

ST-segment elevation myocardial infarction

The two randomized, double-blind, placebo-controlled clinical trials, CLARITY and COMMIT, along with the prespecified subgroup analysis (CLARITY PCI) of the CLARITY study, evaluated the safety and efficacy of clopidogrel in patients with acute ST-segment elevation myocardial infarction (STEMI).

The CLARITY trial enrolled 3,491 patients with ST-segment elevation myocardial infarction (STEMI) occurring within 12 hours who were candidates for thrombolytic therapy. Patients received either clopidogrel (a 300 mg loading dose followed by 75 mg/day, n=1752) or placebo (n=1739), both combined with aspirin (initial loading dose of 150–325 mg followed by 75–162 mg/day), a fibrinolytic agent, and heparin (when appropriate). Patients were followed for 30 days. The primary endpoints included infarction-related arterial occlusion identified on angiography before discharge, or death or recurrent myocardial infarction prior to coronary angiography. For patients who did not undergo angiography, the primary endpoint was death or recurrent myocardial infarction within 8 days or before discharge. The patient population comprised 19.7% women and 29.2% patients aged ≥65 years. Among these patients, the use of fibrinolytic agents (fibrin-specific: 68.7%; non-fibrin-specific: 31.1%) was observed in 99.7% of patients, while heparin was used in 89.5%, β-blockers in 78.7%, ACE inhibitors in 54.7%, and statins in 63%.

The primary endpoint event occurred in 15.0% of patients in the clopidogrel treatment group and 21.7% in the placebo group, indicating that clopidogrel reduced the absolute risk by 6.7% and the relative risk by 36% (95% CI: 24–47%; $p<0.001$), primarily attributed to a significant reduction in arterial occlusion related to infarction. This benefit was consistent across all predefined subgroup analyses, including patient age and gender, infarction site, and type of fibrinolytic agent or heparin used.

The CLARITY PCI subgroup analysis included 1,863 STEMI patients who underwent PCI. Compared with patients receiving placebo (n = 930), those receiving a 300 mg loading dose (LD) of clopidogrel (n = 933) exhibited a significantly lower incidence of post-PCI cardiovascular death, MI, or stroke (clopidogrel pretreatment group: 3.6% vs. placebo group: 6.2%; OR: 0.54; 95% CI: 0.35–0.85; $p = 0.008$). Similarly, patients receiving the 300 mg clopidogrel loading dose showed a significant reduction in the 30-day post-PCI incidence of cardiovascular death, MI, or stroke compared with those receiving placebo (clopidogrel pretreatment group: 7.5% vs. placebo group: 12.0%; OR: 0.59; 95% CI: 0.43–0.81; $p = 0.001$). However, no statistically significant difference was observed when evaluating this composite endpoint in the overall CLARITY study population as a secondary endpoint. No significant difference was noted in the incidence of major or minor bleeding between the two treatment groups (clopidogrel pretreatment...

The rate was 2.0% in the group versus 1.9% in the placebo group (p>0.99). These analytical results support the use of a loading dose of clopidogrel in the early stage of STEMI and also support the strategy of routine clopidogrel pretreatment for patients undergoing PCI.

In the 2×2 factorial design COMMIT trial, a total of 45,852 patients with symptoms suggestive of myocardial infarction (MI) occurring within 24 hours and corresponding electrocardiographic abnormalities (e.g., ST elevation, ST depression, or left bundle branch block) were enrolled. Patients received either clopidogrel (75 mg/day, n=22,961) or placebo (n= 22,891), both combined with aspirin (162 mg/day). Treatment lasted for 28 days or until patient discharge. The primary composite endpoint included death from any cause or the occurrence of reinfarction, stroke, or death. The patient population comprised 27.8% women, 58.4% of whom were aged ≥60 years (26% aged ≥70 years), and 54.5% received fibrinolytic therapy.

Clopidogrel reduced the relative risk of all-cause mortality by 7% (p=0.029) and decreased the relative risk of the composite endpoint of reinfarction, stroke, and death by 9% (p=0.022), with corresponding absolute risk reductions of 0.5% and 0.7%, respectively. This benefit was consistent across age, gender, and whether fibrinolytics were used or not, and was observed as early as 24 hours post-administration.

Patients with acute coronary syndrome who have undergone PCI were administered a loading dose of clopidogrel 600 mg.

CURRENT-OASIS-7*(Using optimized doses of clopidogrel and aspirin to reduce recurrent events in the seventh tissue and evaluate treatment strategies for ischemic syndromes)*

This randomized factorial trial enrolled 25,086 patients with acute coronary syndrome (ACS) scheduled for early percutaneous coronary intervention (PCI). Patients were randomly assigned to receive either a double dose of clopidogrel (600 mg on day 1, 150 mg from days 2–7, followed by 75 mg daily) or a standard dose (300 mg on day 1, followed by 75 mg daily), and compared with high-dose (300–325 mg daily) versus low-dose (75–100 mg daily) aspirin (ASA). Coronary angiography was performed in 24,835 enrolled ACS patients, and PCI was performed in 17,263 patients. Among the 17,263 patients who underwent PCI, the double-dose clopidogrel regimen reduced the incidence of the primary endpoint compared to the standard dose (3.9% vs. 4.5%; adjusted HR = 0.86; 95% CI: 0.74–0.99; p = 0.039) and significantly decreased stent thrombosis (1.6% vs. 2.3%; HR = 0.68; 95% CI: 0.55–0.85; p = 0.001). Major bleeding was more common in the double-dose clopidogrel group than in the standard-dose group (1.6% vs. 1.1%; HR = 1.41; 95% CI: 1.09–1.83; p = 0.009). In this trial, the efficacy of the 600 mg loading dose of clopidogrel was consistent in patients aged ≥75 years and <75 years.

Long-term (12 months) clopidogrel therapy in STEMI patients after PCI

CREDO (*Clopidogrel Reduction of Adverse Events During Observation)*

This study was a randomized, double-blind, placebo-controlled trial conducted in the United States and Canada, aimed at evaluating the benefits of long-term (12 months) clopidogrel therapy following PCI. A total of 2,116 patients were randomly assigned to receive either clopidogrel 300 mg LD (n = 1,053) or placebo (n = 1,063) 3 to 24 hours prior to PCI. All patients also received aspirin 325 mg. Subsequently, all patients in both groups received clopidogrel at a dose of 75 mg/day until day 28. From day 29 through 12 months, patients in the clopidogrel group received clopidogrel at 75 mg/day, while the control group received placebo. Throughout the study, both groups received ASA (81–325 mg/day). At 1 year, the clopidogrel group exhibited a significantly lower composite risk of death, MI, or stroke compared to the placebo group (relative reduction of 26.9%,95% CI: 3.9%–44.4%; p = 0.02; absolute reduction of 3%). At 1 year, no significant increase was observed in the incidence of major bleeding (8.8% vs. 6.7% in the clopidogrel group; p = 0.07) or minor bleeding (5.3% vs. 5.6% in the clopidogrel group; p = 0.84) between the two treatment groups. The primary finding of this study is that continuous treatment with clopidogrel and ASA for at least one year resulted in statistically significant occurrence of major thrombotic events.

The statistical significance and clinical significance have decreased.

Pharmacology and Toxicology

pharmacological action

Clopidogrel is a platelet aggregation inhibitor that selectively inhibits the binding of adenosine diphosphate (ADP) to platelet receptors and the subsequent ADP-mediated activation of the glycoprotein GP*II*b/*III*a complex, thereby suppressing platelet aggregation. Clopidogrel requires biotransformation to exert its anti-aggregant effect. It also blocks platelet aggregation induced by other agonists through ADP release. The action of clopidogrel on platelet ADP receptors is irreversible; consequently, the entire lifespan of platelets exposed to clopidogrel is affected, and the recovery rate of normal platelet function parallels platelet renewal.

Toxicology Study

In preclinical studies conducted in rats and baboons, the most common adverse reaction was hepatic changes. These hepatic alterations resulted from effects on hepatic metabolic enzymes at doses 25 times the human exposure level of 75 mg/day. No significant impact on hepatic metabolic enzymes was observed with therapeutic doses of clopidogrel in humans. Oral administration of high doses of clopidogrel in rats and baboons affected gastric tolerance (gastritis, gastric erosion, and/or vomiting).

genetoxic

No significant abnormalities were observed in either in vivo or in vitro genotoxicity tests of clopidogrel.

Reproductive toxicity

Clopidogrel showed no significant effect on the fertility of either female or male rats, nor did it exhibit notable impacts on the growth and development of offspring in rats and rabbits. Oral administration of clopidogrel to lactating rats resulted in a mild delay in offspring development. Pharmacokinetic studies demonstrated that clopidogrel and/or its metabolites are excreted in breast milk, suggesting a potential direct or indirect effect of clopidogrel.

carcinogenicity

Mice were orally administered clopidogrel for 78 weeks, and rats were orally administered clopidogrel for 104 weeks at a dose of 77 mg/kg. No carcinogenic effects were observed; the plasma concentration at this dose was 25 times higher than the recommended human dose (75 mg/day).

【 Store up 】

No special storage requirements.

【 Pack 】

Dual-aluminum blister packaging, 7 pieces/box, 28 pieces/box, 90 pieces/box

【 Term of Validity 】

36 months.

【 Operative Norm 】

Imported Drug Registration Standard JX20190112.

【 License Number 】

Approval Number for Small Packaging of Imported Drugs: National Drug Approval Number HJ20171237

Approval Number for Large Packaging of Imported Drugs: National Drug Approval Number HJ20171238

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